

Carlos Cardoso

Software Engineer - Rust Backend

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github.com/carlos-cardoso

Summary & Skills

Summary:

I'm a Rust Engineer with a background in data processing and distributed systems. Skilled in delivering high-performance software projects from inception to completion, with a deep understanding of Rust and computer architecture. I also have seven years of research experience in the field of machine learning and robotics at VisLab, Computer Vision Laboratory. I authored first-author publications in IEEE conferences, including ICRA, ICDL, and IROS, giving me a strong math background, and excellent communication skills.

I have a self-starter mentality; In my current position in an AI startup I took the initiative in the conversion of microservices to use CQRS, rethought the way we handle patient metadata in one of our systems, and improved our encryption protocols. In addition to startup and research experience I have done technical consulting for artists in Berlin and Lisbon, where I had to adapt to tight deadlines and quickly changing requirements ([projects](#)).

Skills

- **Languages & Systems:** Rust, C/C++, Python, Kubernetes, Unix, bash, Git, Docker, TypeScript, SQL, Nix
- **Distributed Systems:** CQRS, Event Sourcing, Kubernetes, Micro services
- **Tools & Frameworks:** Knative, PostgreSQL, Pytorch, Polars, Deep Learning, Apache Datafusion, Backstage

Experience

Rust, Distributed Systems, AI, Event Sourcing, CQRS, Kubernetes, Knative, PostgreSQL, Encryption Protocols

Rust Engineer - deepc.ai

January 2023 | Present, Munich, Germany

- Contributed to the architecture of the on-premise and cloud kubernetes clusters that integrate with the radiology machines and medical imaging visualizers (PACS) across a suite of diagnostic modalities (Ultrasound, MMG, SPECT, PET, MRI, CT, X-Ray) for DeepcOS, a cloud-based AI platform that enhances the diagnostic capabilities of radiologists globally.
- Orchestrated data preprocessing and pseudonymization workflows, ensuring secure and efficient data routing to multiple AI engines for medical image analysis.
- Implemented CQRS and Event Sourcing patterns within microservices, contributing to a robust architecture capable of maintaining performance under heavy loads.
- Demonstrated with proof of concepts and fixed vulnerabilities in our encryption protocols.
- Collaborated with a team with a strong Software Engineering ethos (automated CI/CD, code reviews, pair programming), fostering personal growth and a shared enthusiasm for AI in healthcare innovation.

C++, Python, Embedded,
Bare metal, Behavior Trees,
Sensor Fusion

Rust, C++, Python, Statistical Learning,
Deep Learning, Computer Vision,
Pytorch, Tensorflow

Teaching, Technical Writing,
Low Level, Embedded
Javascript, Distributed Control

Embedded Software Engineer - Feerica S.A.

January 2016 - August 2016; March 2022 - December 2022 Mafra, Portugal / Remote

- Developed firmware for secure cash-in-transit systems, focusing on integration with ARM microprocessors, GPS, and other embedded peripherals.
- Implemented behavior trees for customizing embedded business logic, and optimized accelerometer signal processing for fraud detection in cash transit systems.
- Collaborated with hardware engineers to design and implement low-level device drivers, including custom logic for security systems, aligning with cryptographic requirements for secure data handling.

Robotics Researcher - VisLab ISR

October 2016 - March 2022, Lisbon, Portugal

The Robot Vision Laboratory (VisLab) of ISR/LarSYS is a top graded research institution in Portugal. I researched Machine Learning applied to Physical Human-Robot Interaction, focusing on tasks such as Human-Robot handover and handshaking. I published papers as a first author in top IEEE conferences (see [publications](#)). In addition, I supervised MSc and BSc students in robotics and deep learning projects and contributed to **open-source projects in C++, Python**. Postgraduate Courses: **Convex Optimization** (19/20); **Deep Structured Learning** (17/20); **Statistical Learning** (17/20); Outreach and Teaching Skills (17/20).

Teaching Assistant (Real-Time Distributed Control Systems) | Técnico Lisboa - University of Lisbon

September 2017 - February 2018, Lisbon, Portugal

I worked as a TA for the MSc. level course on Real-Time Distributed Control Systems (SCDTR). Created an evaluation sheet generator that fetches student information from the university's database.

Helped students with different backgrounds (Electrical Eng. and Aerospace Eng.) control lights according to a **distributed** system of luminance sensor readings (ADMM method running on Microcontrollers). Evaluated the student's final projects.

Education

Técnico Lisboa / BSc and MSc in Electrical & Computer Engineering

April 2016, Lisbon, Portugal

MsC Thesis: Reinforcement Learning for a Ping Pong Playing Robot. Grade 18/20

IIT: Instituto Italiano di Tecnologia/ iCub Summer School

July 2015, Sestri Levante, Italy

Örebro University/ Lucia AI & Robotics Winter School

December 2016, Örebro, Sweden

Research Projects/Grants

European Research Council Advanced Grant "ORIENT"

November 2020 - November 2021

University of Lisbon - Fully Funded PhD. Scholarship

October 2017 - October 2020

ISR/LARsYS Strategic Funds from FCT [UID/EEA/5009/2013]

October 2016 - October 2017

European Project: POETICON++ Robots Need Language

December 2014 - December 2015